

Overlapping tiles

To the naked eye, the wings of this peacock butterfly look smooth and velvety.

However, closer examination under a microscope reveals a covering of tiny scales, which are arranged in overlapping rows, like tiles on a roof. A

butterfly's scales are flat and hollow, and they give the wings their magnificent color.



Background wing color produced by pigments in scales

Iridescent eye spots created by reflective scales

Tiny ridges on surface of scales reflect light, producing different colors

Microscopic view of butterfly scales

Head scales caked with salt sprayed from glands near eyes



Scary scales

The marine iguana has large scales on its head that form different shapes, such as knobs and cones, and a crest of spike-shaped scales running down its back. Despite its fearsome appearance, it is a harmless vegetarian, feeding entirely on seaweed.

Slippery scales

Snakes have smooth scales, arranged in diagonal rows. Their belly scales are larger and set in horizontal rows to help them slide across the ground. Other reptiles, including crocodiles, tortoises, and some lizards, have armored, bony scales called scutes. In giant tortoises, shell scutes can be nearly 10 in (25 cm) across.

Snake's eye protected by a single, clear scale, or spectacle

